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# The nexus of managerial and technical AI knowledge, disruptive innovation and the circular economy: The role of organizational change capability and financial resilience

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ABSTRACT

Drawing on the dynamic capabilities theory, this study investigates how managerial and technical artificial intelligence (AI) knowledge influence disruptive innovation and its impact on the circular economy. It also examines the moderating effects of organizational change capability and financial resilience on the relationship between disruptive innovation and the circular economy. The proposed model and its associated hypotheses were tested using Partial Least Squares (PLS) structural equation modeling (SEM). This study is based on two-wave data collected from 242 general, IT, and operations managers in firms located within a central hub for the technology and manufacturing industries in Baghdad. The findings indicate that managerial and technical AI knowledge significantly boost disruptive innovation, which, in turn, enhances the circular economy. Moreover, organizational change capability and financial resilience strengthen the relationship between disruptive innovation and the circular economy. This research contributes to the field of AI capabilities by highlighting their role in fostering disruptive innovation and promoting sustainability through the circular economy. Additionally, it provides insights into how organizational and financial resilience, alongside AI skills and knowledge, can support sustainable business models. Practically, the study underscores the necessity for technology and manufacturing companies to invest in both managerial and technical AI skills while prioritizing robust organizational change capabilities and financial resilience to maximize the benefits of disruptive innovation and support a circular economy.

## 1. Introduction

Businesses are increasingly adopting the circular economy in response to growing pressures from resource constraints and regulatory demands (Khan et al., 2024). Resource scarcity poses significant challenges in multiple industries as finite resources such as minerals, metals, and fossil fuels face depletion, leading to price volatility and supply chain disruptions. This makes the traditional linear model of take, make, and dispose economically unsustainable and strategically risky. Circular economy enables firms to optimize resource use and reduce reliance on limited inputs (Castro-Lopez et al., 2023; Guo & Tsinopoulos, 2023; Mishra et al., 2023). Moreover, governments are implementing stricter environmental policies and sustainability targets due to the climate crisis, compelling businesses to reevaluate their operational models. Carbon taxes, extended producer responsibility (EPR) frameworks, and waste reduction mandates drive companies to align their strategies with sustainability objectives (Al Halbusi et al., 2025; Tiwari et al., 2023).

In addition to these pressures, consumer demand for environmentally responsible products and services has surged. Today's consumers are aware of the environmental footprint of their purchases and increasingly favour brands that demonstrate a commitment to sustainability (Sahoo, 2024; Tiwari et al., 2023). Businesses embracing circular economy practices can improve their brand image, attract eco-conscious customers, and differentiate themselves in competitive markets. Moreover, adopting circular economy can drive significant cost savings by reducing material waste, increasing resource efficiency, and fostering

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innovation (Du et al., 2023; Tiwari, 2023; Zhang et al., 2023). Circular practices such as remanufacturing, recycling, and using renewable energy sources offer firms opportunities to diversify revenue streams (Milios et al., 2019; Patel and Singh, 2023), develop new business models, and gain a competitive edge. By addressing resource pressures, complying with regulatory requirements, and meeting consumer expectations, companies that transition to a circular economy model achieve greater sustainability and position themselves for long-term growth and resilience (Ayaz & Tatoglu, 2024; Tiwari et al., 2023).

The circular economy is a transformative approach aimed at decoupling economic growth from resource consumption while promoting environmental sustainability and business efficiency. In contrast to the traditional linear economy (characterized by extracting, consuming, and disposing of resources) circular economy is founded on three core principles: reducing waste, reusing materials, and recycling resources (Hossain et al., 2024; Patel and Singh, 2023). This model emphasizes designing products, processes, and systems that preserve material value throughout their lifecycle, minimizing waste and enhancing resource efficiency. It boosts innovation in product design, focusing on modularity, reparability, and recyclability to extend product lifespans and enable secondary material use (Du et al., 2023).

Circular economy highlights the importance of industrial collaboration in establishing closed-loop systems, where waste from one process becomes input for another (Bhatia et al., 2024; Guo & Tsinopoulos, 2023; Mishra et al., 2023). For instance, manufacturing, construction, and technology companies are forming innovative partnerships to reuse byproducts, minimize landfill waste, and enhance resource sharing. This approach not only addresses environmental issues but also generates economic benefits by reducing production costs and creating new revenue opportunities, such as leasing models, remanufactured products, and resource-sharing platforms (Baldassarre et al., 2019).

Circular economy supports sustainability goals by addressing climate change and preserving biodiversity. It reduces reliance on virgin resources and minimizes greenhouse gas emissions, aligning with global initiatives like the Paris Agreement. Circular economy also fosters social sustainability through the creation of jobs in recycling, repair, and sustainable design. For businesses, adopting circular economy practices boosts operational efficiency, enhances resilience to resource volatility, and reinforces corporate social responsibility. Consequently, the circular economy is increasingly essential for a sustainable future that harmonizes economic growth with environmental and social well-being (Arranz et al., 2023; Gallego-Schmid et al., 2020).

Enhancing the circular economy requires incorporating key components that foster sustainable and resource-efficient systems. Innovative design plays a pivotal role by prioritizing the creation of durable, repairable, and recyclable products, thereby delaying the circulation of materials within the economy. Resource efficiency is equally critical, emphasizing optimized raw material utilization while minimizing waste through strategies such as upcycling, reusing, and reducing resource consumption (Vann Yaroson et al., 2024).

Disruptive technologies, including Artificial Intelligence (AI), enable advancements such as real-time material tracking, predictive maintenance, and improved efficiency (Arora et al., 2023; Soto-Acosta, 2020, 2024). Furthermore, policy and regulatory frameworks provide critical support to circular economy by incentivizing sustainable practices and imposing restrictions on linear economic activities, such as landfilling and single-use plastics (Ajayi & Udeh, 2024). Collaborative partnerships among businesses, governments, and communities further enhance circular economy by promoting innovation and fostering shared responsibility in resource management. Consumer awareness is also fundamental, as informed and sustainability-conscious individuals drive demand for circular economy practices, including purchasing refurbished goods and participating in recycling initiatives. Financial mechanisms, such as green financing, subsidies, and tax incentives, play a vital role in enabling businesses to transition toward circular practices by alleviating the burden of initial investment costs. By integrating these interconnected components, organizations and societies can develop a comprehensive framework for achieving circular economy objectives, advancing environmental sustainability, economic resilience, and societal well-being (Kutaula et al., 2022).

Prior studies reveal a significant lack of understanding regarding how managerial and technical AI skills contribute to disruptive innovation, particularly in transitioning to a circular economy. Most existing literature examines these AI competencies separately, leaving their combined impact on sustainability-driven innovation underexplored (e.g., Scuotto et al., 2022). Moreover, the link between disruptive innovation and circular economy is often considered critical, as innovation is essential for promoting sustainable practices such as resource optimization, waste reduction, and circular business models. Importantly, the role of organizational change capability, which enables firms to adapt and reconfigure resources in response to disruption, has been inadequately addressed in circular economy research. This gap limits understanding of how organizational agility and strategic flexibility can enhance the adoption of circular economy principles (Cegarra-Navarro et al., 2016). Furthermore, financial resilience (Arana-Landin et al., 2024; Zahedi et al., 2023), a crucial yet often neglected outcome of circular economy, is rarely explored, as most studies focus on environmental and operational benefits without considering the economic stability circular economy can offer.

This study aims to fill these gaps by proposing a comprehensive framework that integrates managerial and technical AI skills, positions disruptive innovation as a key link to circular economy, emphasizes the role of organizational change capability, and frames financial resilience as a critical boundary condition—both an enabler and an outcome of circular economy. Thus, this study makes critical contributions to theoretical understanding while providing practical insights for firms seeking to align their innovation strategies with long-term sustainability and economic viability. This holistic approach addresses key gaps in the literature and offers actionable pathways for sustainable business transformation.

## 2. Theory and hypotheses development

Dynamic Capabilities Theory (Teece et al., 1997) explains how organizations develop and adapt their capabilities to thrive in changing environments, as illustrated in Fig. 1. Managerial and technical AI skills are essential for identifying opportunities and leveraging disruptive innovation to gain competitive advantages. Disruptive innovations reshape traditional market structures by offering more efficient, scalable, and sustainable solutions, necessitating substantial changes in resource allocation and business processes. Organizational change capability plays a critical role in implementing these changes effectively as it enables firms to manage resistance, reconfigure workflows, and align stakeholders with shared goals (Cegarra-Navarro et al., 2016). Furthermore, these innovations contribute to broader objectives such as the circular economy by promoting resource optimization, waste reduction and sustainable value creation—priorities for policymakers and consumers alike. In addition, dynamic capabilities enhance

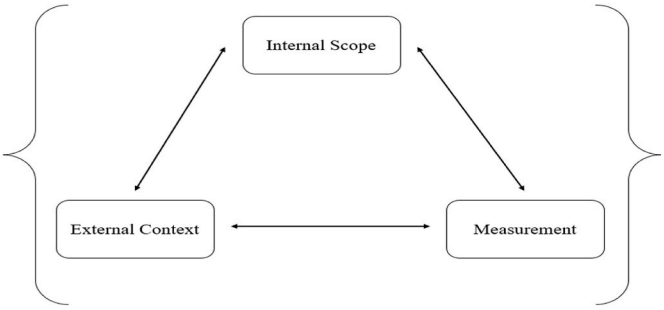



**Fig. 1.** Dynamic capabilities theory in context.

financial resilience, enabling organizations to navigate uncertainties, mitigate risks, and seize new revenue opportunities (Rotjanakorn et al., 2020; Zahedi et al., 2023). This framework highlights how businesses can integrate AI-driven skills and knowledge, innovation, and adaptability to build enduring resilience and sustainability while remaining competitive in rapidly evolving markets.

### 2.1. Managerial AI skills and disruptive innovation

Managerial AI skills are essential for organizations aiming to harness artificial intelligence for innovation, improved decision-making, and competitive advantage. These skills enable managers to integrate AI into business strategies, optimize processes (Chowdhury et al., 2023), and identify new opportunities, allowing organizations to adapt to technological changes and leverage emerging market trends.

The relationship between managerial AI skills and disruptive innovation has garnered significant attention in recent literature, emphasizing the transformative potential of managerial expertise in leveraging AI technologies to drive innovation. Managerial AI skills, which encompass the ability to understand, interpret, and strategically apply AI-driven insights, are increasingly recognized as critical enablers of disruptive innovation (Sousa & Rocha, 2019). These skills empower managers to identify emerging trends, automate complex decision-making processes, and reconfigure business models to achieve competitive advantage (Saurabh et al., 2022).

Researchers argue that managers proficient in AI can effectively bridge the gap between technical AI capabilities and strategic business objectives, fostering environments conducive to breakthrough innovations. Moreover, integrating AI into managerial decision-making processes facilitates the identification of inefficiencies (Brink et al., 2024), the prediction of market shifts, and the development of innovative products or services that challenge industry norms.

The literature also highlights that managerial AI skills enhance organizational adaptability, enabling proactive responses to technological disruptions and capitalizing on new market opportunities (Baabdullah, 2024). However, despite these potential benefits, challenges such as the need for continuous upskilling, ethical considerations, and organizational resistance to change can hinder the effective deployment of AI for disruptive innovation.

Overall, aligning managerial AI skills with strategic goals is pivotal in fostering disruptive innovation, transforming industries, and sustaining competitive advantage in an increasingly AI-driven business landscape (Sullivan & Wamba, 2024). The following hypothesis reflects these expectations:

**Hypothesis 1.** There is a positive relationship between managerial AI skills and disruptive innovation

### 2.2. Technical AI skills and disruptive innovation

Technical AI skills are crucial for developing, implementing, and optimizing artificial intelligence solutions within organizations (Olan et al., 2022). These skills empower professionals to design and deploy AI systems, analyze complex data sets, and create algorithms that foster automation and innovation. Mastery of these skills enables organizations to fully leverage AI, enhance efficiency, address complex challenges, and generate data-driven solutions.

Recent research highlights the connection between technical AI skills and disruptive innovation, as organizations increasingly acknowledge the critical role of advanced AI capabilities in driving technological advancements (Wael Al-Khatib, 2023). Expertise in machine learning, data analytics, neural networks, and algorithm development is essential for creating and implementing AI-driven innovations that disrupt industries. Studies reveal that organizations with robust technical AI capabilities are better positioned to develop new products, services, and business models that challenge traditional market structures (Amankwah-Amoah et al., 2024).

These skills facilitate the design of AI solutions that enhance operations, improve customer experiences, and introduce transformative technologies. Furthermore, technical AI expertise enables firms to quickly identify and respond to emerging trends, strengthening their capacity to leverage disruptive innovations (Parise et al., 2016). However, the literature emphasizes that the full potential of technical AI skills is realized only when effectively integrated with business strategy and aligned with organizational objectives (Chowdhury et al., 2022). Thus, the following hypothesis is posited:

**Hypothesis 2.** There is a positive relationship between technical AI skills and disruptive innovation

### 2.3. Disruptive innovation and circular economy

Disruptive innovation is crucial for driving transformative changes across industries, as it introduces new technologies, business models, or processes that fundamentally alter market dynamics. By challenging established practices, disruptive innovation fosters increased efficiency, enhances consumer experiences, and creates new growth opportunities, enabling organizations to remain competitive and adapt to evolving market demands (Schindler et al., 2024).

The relationship between disruptive innovation and the circular economy has garnered significant attention in recent academic literature as organizations transition from linear to sustainable business models (Liu et al., 2024). Disruptive innovation introduces technologies or business models that transform market dynamics, playing a crucial role in advancing the circular economy. Scholars argue that innovations such as digital technologies, artificial intelligence, and advanced manufacturing can help organizations reduce waste, optimize resource use, and close product lifecycles, key principles of the circular economy (Popa et al., 2022; Saha et al., 2024). These innovations enable more efficient recycling systems, product redesigns for reuse and repair, and sustainable business models that prioritize environmental impact over traditional growth metrics. Furthermore, disruptive innovations help overcome technical and economic barriers that have historically hindered circular practices, facilitating the adoption of circular business models. The literature also highlights the potential of these innovations to influence consumer behaviour, encouraging sustainable choices and further promoting the transition to a circular economy (Pascucci et al., 2024). However, challenges such as regulatory constraints, technological limitations, and market resistance persist. Integrating disruptive innovation with circular economy principles presents a transformative opportunity to enhance environmental sustainability and economic resilience (Sehnem et al., 2022). Based on this, the following hypothesis is proposed:

**Hypothesis 3.** There is a positive relationship between disruptive innovation and the circular economy

### 2.4. The role of organizational change capability

Organizational change capability is vital for firms to effectively navigate and adapt to transformations such as technological advancements or market shifts. It fosters a culture of adaptability, resilience, and innovation, which is essential for maintaining competitiveness, driving continuous improvement, and successfully implementing new strategies, business models, or practices (Joel et al., 2024; Purkayastha and Gupta, 2023).

The current study identifies organizational change capability as a moderating factor in the relationship between disruptive innovation and the circular economy, as it significantly influences how effectively an organization can embrace and implement the changes driven by these transformative forces (Supriharyanti & Sukoco, 2023). Disruptive innovation often leads to radical shifts in business models, technologies, and processes, creating potential organizational resistance. Similarly,

transitioning to a circular economy requires substantial changes in operational practices, such as redesigning products for reuse, rethinking supply chains (Bansal et al., 2023), and adopting sustainable practices throughout the value chain.

Organizations with strong change capabilities are better equipped to manage these disruptions by fostering a culture of adaptability, resilience, and continuous learning, which enables them to integrate new technologies and practices seamlessly. Firms with high change capability can ensure smoother transitions when adopting innovative business models, technologies, or sustainability initiatives, thus reducing friction and accelerating the successful integration of disruptive innovation and circular economy principles (Shaik et al., 2024). These organizations are more likely to align their strategic objectives with evolving market demands, driving sustainable growth.

Conversely, firms with limited change capacity may struggle with these transitions, resulting in slower adoption, inefficiencies, or resistance to new paradigms. Therefore, organizational change capability serves as a critical moderator, facilitating the alignment of disruptive innovations with the strategic goals of a circular economy and ensuring effective implementation. This alignment leads to long-term sustainability, competitiveness, and organizational success (Kuhlmann et al., 2023).

Based on these arguments, the following hypothesis is posited:

**Hypothesis 4**. The positive relationship between disruptive innovation and the circular economy is moderated by organizational change capability such that this relationship will be stronger when organizational change capability is high.

### 2.5. The role of financial resilience

Financial resilience is crucial for organizations to withstand economic challenges, absorb financial shocks, and continue investing in growth and innovation. It enables firms to maintain stability during times of uncertainty (Pal et al., 2014; Zahedi et al., 2023), adapt to market changes, and fund long-term strategic initiatives, ensuring sustainable success despite disruptions or unexpected setbacks.

Financial resilience plays a significant moderating role in the relationship between disruptive innovation and the circular economy, as it enables organizations to withstand the financial challenges and uncertainties associated with these transformative changes (Karman et al., 2024). Disruptive innovation often requires significant upfront investments in new technologies, research, and business model redesign. Simultaneously, the transition to a circular economy involves rethinking production processes, investing in sustainable materials, and adopting waste-reducing practices. Organizations with strong financial resilience are better equipped to absorb these costs and manage the risks associated with these transitions (Miroshnychenko et al., 2024). Financial resilience allows firms to maintain stable cash flows, access necessary capital, and weather short-term financial setbacks, ensuring they can continue investing in disruptive innovations and circular economy practices without compromising their long-term sustainability.

Moreover, financially resilient organizations are more likely to take calculated risks and make strategic investments that align with innovation and sustainability goals. In contrast, firms with weak financial resilience may struggle to secure the funding required for such investments (Carmeli et al., 2020), which could hinder their ability to capitalize on disruptive innovations or fully transition to circular economy models. Therefore, financial resilience serves as a critical moderator, facilitating the successful implementation of disruptive innovation and circular economy practices by providing the financial stability necessary to support these long-term, transformative changes (Lu et al., 2022). Thus, the following hypothesis is proposed:

**Hypothesis 5**. The positive relationship between disruptive innovation and the circular economy is moderated by financial resilience such that this relationship will be stronger when financial resilience capability is high.

### 2.6. Research model

This study develops an integrative research model to explore the influence of managerial and technical AI skills on disruptive innovation and its subsequent impact on the circular economy, grounded in dynamic capabilities theory. This theoretical framework underscores how organizations in Baghdad, Iraq's technology and manufacturing sectors adapt, integrate, and reconfigure resources to address rapidly evolving challenges and opportunities. By emphasizing AI-driven competencies, the model examines the pathways through which technological and managerial capabilities foster disruptive innovation aligned with circular economic principles.

Additionally, the moderating roles of organizational change capability and financial resilience are incorporated into the research model, providing valuable insights into how adaptability and resource stability enhance the effectiveness of disruptive innovation. The network of relationships proposed in this study is presented in Fig. 2. This approach offers a robust foundation for understanding how organizations in these critical sectors can strategically leverage AI skills to achieve sustainable growth and resilience within the framework of a circular economy.

## 3. Methodology

This study employed a quantitative approach to investigate the relationships between managerial AI skills, technical AI skills, disruptive innovation, organizational change capability, financial resilience, and circular economy practices within Iraq's technology and manufacturing sectors. A structured survey methodology was utilized to collect data, ensuring a clear focus on key variables and enabling the quantification of relationships. The survey design aimed to provide comprehensive insights into how AI integration and innovation influence organizational performance in Iraq's evolving industrial landscape.

### 3.1. Sample

Data were gathered from a sample of key decision-makers, including general managers, IT managers, and operations managers, working in firms located in Baghdad, a central hub for Iraq's technology and manufacturing industries. These sectors were chosen for their critical role in driving economic development and their potential for transformation through innovation and sustainability practices.

The sample was strategically selected to include companies of various sizes and operational structures, providing diverse perspectives on the challenges and opportunities in Baghdad's technology and manufacturing sectors. By focusing on key managerial roles, the research captured insights into leadership's influence on driving change and technology adoption, particularly in the context of Iraq's recovery and modernization. This sampling approach was designed to enhance the study's validity and reliability, offering a robust basis for evaluating the sectors' readiness for innovation and sustainability.

A total of 242 valid surveys were completed. To mitigate common method bias, data were collected from different respondents over two time periods (Podsakoff et al., 2024). In the first survey, IT managers assessed managerial and technical AI skills. In the second survey, general and operations managers evaluated disruptive innovation, organizational change capability, financial resilience, and circular economy practices. Of the 265 distributed questionnaires, 249 responses were received, and 242 were deemed valid after excluding seven due to irregularities, resulting in a 95 % response rate. Both survey sets were coded to match responses across the two time periods, further enhancing the robustness of the findings.

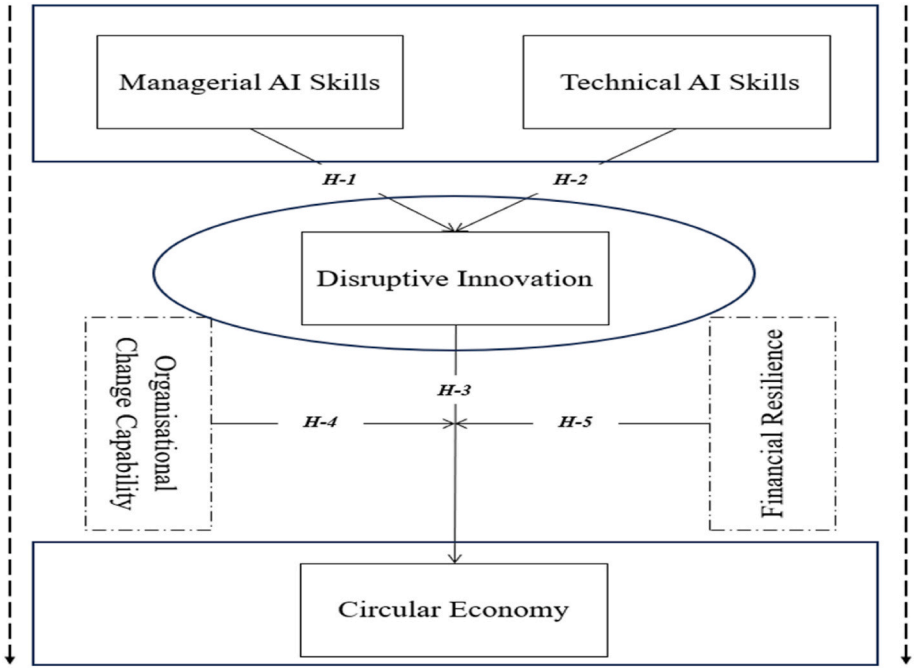



**Fig. 2.** Research model.

### 3.2. Variables measurement

The variables were measured using scales adapted from previous studies to ensure validity and reliability. AI managerial and technical skills were assessed using seven items each, based on measures from Mikalef and Gupta (2021) and An et al. (2024). Disruptive innovation was measured with three items adapted from Kraus et al. (2023), while organizational change capability was evaluated using eight items from Roux et al. (2023). Financial resilience and circular economy practices were each assessed using ten items, adapted respectively from Gupta et al. (2023) and Al Halbusi et al. (2024, 2025). Control variables, including firm size (e.g., number of employees and annual revenue), firm age, market competition, R&D expenditure, and technology adoption stage, were also included to account for potential confounding effects on performance and innovation.

Since the study was conducted in an Arabic-speaking context, survey items were translated into Arabic using the back-translation method (Brislin, 1980), a widely recognized approach for ensuring accuracy in cross-cultural surveys. The initial translation was followed by a comparison between the Arabic and original English versions by two professional bilingual speakers to identify discrepancies. Any differences were resolved through retranslation and further evaluation until the experts confirmed semantic equivalence, achieving a high agreement rate.

Before administering the questionnaire, the psychometric properties of the scale items were thoroughly evaluated. Face and content validity testing involved input from experts in business and operations management, PhD students, and information management specialists (Memon et al., 2017).

The questionnaire, summarized in Table 1, used a five-point Likert scale ranging from "1 = Strongly Disagree" to "5 = Strongly Agree" to capture responses across constructs central to the study. This robust design ensured the accuracy and reliability of data collection for the subsequent analysis.

## 4. Data analysis and results

Structural Equation Modeling (SEM) using Partial Least Squares (PLS) was employed to analyze the study's research model, utilizing Smart PLS 4 software (Ringle et al., 2015). This robust statistical method (Hair et al., 2017) does not impose strict distributional assumptions, making it particularly suitable for complex causal analyses involving first- and second-order constructs, as in this study. Following established guidelines (Hair et al., 2017), the PLS analysis was conducted with 5000 subsamples to compute bootstrap t-statistics with n–1 degrees of freedom (where n is the number of subsamples) to test path coefficient significance. The PLS approach follows a two-stage process: first, examining the measurement model for scale validity and reliability, and second, analyzing the structural model to test the hypotheses.

The measurement model was assessed by analyzing individual item reliability, internal consistency, convergent validity, and discriminant validity. The item reliability showed no significant issues, with most items exceeding the recommended 0.707 threshold (Hair et al., 2017). Composite reliability (CR) for internal consistency was confirmed, with all values surpassing the 0.70 cutoff (Hair et al., 2017). Specifically, Cronbach's Alpha, Dillon-Goldstein (ρc), and Dijkstra-Henseler (ρA) values all exceeded 0.70 (Hair et al., 2017). For convergent validity, the average variance extracted (AVE) values confirmed validity by exceeding the 0.5 threshold (Hair et al., 2017).

The Fornell-Larcker method was used to assess discriminant validity, and no issues were identified, as the AVE for each construct exceeded the shared variance with other latent variables (see Table 2) (Hair et al., 2017). Additionally, the Heterotrait-Monotrait Ratio (HTMT), a more robust alternative to Fornell-Larcker, was applied. As shown in Table 3, all HTMT values were below the 0.90 threshold, confirming discriminant validity among variable pairs. Furthermore, the 95 % confidence intervals for HTMT values did not include 1, further supporting discriminant validity (Henseler et al., 2015).

The structural model evaluation followed Hair et al.'s (2017) recommendations, including assessing collinearity, t-values, and bootstrapping procedure with 5000 re-samples. While p-values indicate the presence of effects, they do not measure their magnitude, which is reflected in the path coefficients.

The direct hypothesis testing (H1–H3) revealed the following results. Managerial AI skills significantly predicted disruptive innovation (H1), with $\beta = 0.38$, $t = 6.46$, and $p < 0.000$, providing support for H1. Similarly, the second direct effect (H2) indicated a significant positive relationship between technical AI skills and disruptive innovation, with

**Table 1**
Item loadings construct reliability and convergent validity.

| Construct | Item | VIF | Loading | Construct Reliability | | | AVE |
|---|---|---|---|---|---|---|---|
| | | | | Cronbach's Alpha | Dillon-Goldstein (ρc) | Dijkstra-Henseler (ρA) | |
| Managerial AI Skills | MAIS1 | 1.10 | 0.789 | 0.753 | 0.877 | 0.844 | 0.641 |
| | MAIS2 | 1.22 | 0.881 | | | | |
| | MAIS3 | 2.16 | 0.844 | | | | |
| | MAIS4 | 1.41 | 0.910 | | | | |
| | MAIS5 | 1.21 | 0.812 | | | | |
| | MAIS6 | 1.11 | 0.884 | | | | |
| | MAIS7 | 1.18 | 0.835 | | | | |
| Technical AI Skills | TAIS1 | 1.27 | 0.883 | 0.787 | 0.874 | 0.854 | 0.597 |
| | TAIS2 | 1.46 | 0.819 | | | | |
| | TAIS3 | 1.31 | 0.844 | | | | |
| | TAIS4 | 1.51 | 0.787 | | | | |
| | TAIS5 | 1.17 | 0.866 | | | | |
| | TAIS6 | 1.24 | 0.841 | | | | |
| | TAIS7 | 1.36 | 0.782 | | | | |
| Disruptive Innovation | DIN1 | 1.10 | 0.853 | 0.811 | 0.874 | 0.837 | |
| | DIN2 | 2.11 | 0.871 | | | | |
| | DIN3 | 1.32 | 0.861 | | | | |
| Organizational Change Capability | OCC1 | 1.32 | 0.837 | 0.822 | 0.867 | 0.837 | 0.617 |
| | OCC2 | 1.16 | 0.861 | | | | |
| | OCC3 | 2.04 | 0.823 | | | | |
| | OCC4 | 1.21 | 0.787 | | | | |
| | OCC5 | 1.37 | 0.907 | | | | |
| | OCC6 | 1.19 | 0.831 | | | | |
| | OCC7 | 1.81 | 0.794 | | | | |
| | OCC8 | 1.21 | 0.866 | | | | |
| Financial Resilience | FR1 | 0.142 | 0.791 | 0.788 | 0.851 | 0.817 | 0.601 |
| | FR2 | 0.211 | 0.884 | | | | |
| | FR3 | 0.182 | 0.847 | | | | |
| | FR4 | 0.193 | 0.857 | | | | |
| | FR5 | 0.172 | 0.844 | | | | |
| | FR6 | 0.102 | 0.774 | | | | |
| | FR7 | 0.135 | 0.787 | | | | |
| | FR8 | 0.138 | 0.814 | | | | |
| | FR9 | 0.191 | 0.819 | | | | |
| | FR10 | 0.121 | 0.852 | | | | |
| Circular Economy | CE1 | 0.117 | 0.787 | 0.814 | 0.881 | 0.826 | 0.611 |
| | CE2 | 1.12 | 0.818 | | | | |
| | CE3 | 2.39 | 0.874 | | | | |
| | CE4 | 2.12 | 0.798 | | | | |
| | CE5 | 0.21 | 0.776 | | | | |
| | CE6 | 1.18 | 0.827 | | | | |
| | CE7 | 1.16 | 0.814 | | | | |
| | CE8 | 1.25 | 0.827 | | | | |
| | CE9 | 1.21 | 0.814 | | | | |
| | CE10 | 0.18 | 0.877 | | | | |

**Notes:** AVE = Average Variance Extracted.

**Table 2**
Analysis of Discriminant Validity using Fornell and Larcher

| Constructs | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1. Managerial AI Skills | ***0.774*** | | | | | |
| 2. Technical AI Skills | 3.45 | ***0.687*** | | | | |
| 3. Disruptive Innovation | 4.47 | 0.321 | ***0.767*** | | | |
| 4. Organizational Change Capability | 4.58 | 0.432 | 0.232 | ***0.665*** | | |
| 5. Financial Resilience | 4.68 | 0.521 | 0.335 | 0.383 | ***0.812*** | |
| 6. Circular Economy | 4.34 | 0.532 | 0.411 | 0.514 | 0.551 | ***0.847*** |

**Table 3**
Analysis of discriminant validity using HTMT.

| Constructs | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1. Managerial AI Skills | | | | | | |
| 2. Technical AI Skills | 0.342 | | | | | |
| 3. Disruptive Innovation | 0.317 | 0.344 | | | | |
| 4. Organizational Change Capability | 0.471 | 0.404 | 0.524 | | | |
| 5. Financial Resilience | 0.181 | 0.261 | 0.371 | 0.558 | | |
| 6. Circular Economy | 0.284 | 0.614 | 0.267 | 0.293 | 0.149 | |

Notes: HTMT should be lower than 0.85.

$\beta = 0.198$, $t = 3.11$, and $p < 0.001$, supporting H2. For H3, a significant relationship was observed between disruptive innovation and the circular economy, with, with $\beta = 0.280$, $t = 3.99$, and $p < 0.000$, supporting H3. Results are presented in Table 4.

The moderation test examined whether organizational change capability and financial resilience influenced the relationship between disruptive innovation and the circular economy. Bootstrapping analysis revealed that the interaction between disruptive innovation and organizational change capability significantly affected the circular economy ($\beta = 0.231$, $t = 4.64$, and $p < 0.000$), supporting H4. Similarly, the interaction between disruptive innovation and financial resilience showed significance ($\beta = 0.149$, $t = 2.98$, and $p < 0.000$), providing support for H5.

The findings also indicated varying impacts of firm characteristics on circular economy practices. Firm size exhibited a non-significant relationship ($\beta = -0.04$, $t = 0.70$, suggesting that operational scale has limited influence on the transition to a circular economy. This could stem from structural challenges in larger firms or resource limitations in

**Table 4**
Structural path analysis: Direct effect.

| Hypothesis | Relationship | SB | SD | t-value | p-values | Bias and Corrected Bootstrap 95 % CI [Lower Level; Upper Level] | Decision |
|---|---|---|---|---|---|---|---|
| H-1 | Managerial AI Skills- > Disruptive Innovation | 0.386 | 0.060 | 6.46 | 0.000 | [0.275; 0.475] | Significant |
| H-2 | Technical AI Skills- > Disruptive Innovation | 0.198 | 0.066 | 3.11 | 0.001 | [0.075; 0.293] | Significant |
| H-3 | Disruptive Innovation- > Circular Economy | 0.280 | 0.070 | 3.99 | 0.000 | [0.166; 0.404] | Significant |
| **Analysis of Two-Way Interaction (Moderation)** | | | | | | | |
| H-4 | Disruptive Innovation **X** Organisational Change Capability- > Circular Economy | 0.231 | 0.057 | 40.64 | 0.000 | [0.042; 0.271] | Significant |
| H-5 | Disruptive Innovation **X** Financial Resilience- > Circular Economy | 0.149 | 0.052 | 20.98 | 0.000 | [0.064; 0.274] | Significant |

Notes: n = 242. Bootstrap sample size = 5000. SE = standard error; LL = lower limit; CI = confidence interval; UL = upper limit 95 % bias-correlated CI.

smaller firms. Firm age demonstrated a significant positive effect (β = 0.19, t = 2.73, likely due to accumulated experience, knowledge, and stakeholder relationships facilitating sustainability adoption. Market competition positively influenced circular economy adoption (β = 0.24, t = 3.32), reflecting the drive for innovation in competitive environments. Technology adoption stage exhibited the strongest positive relationship (β = 0.301, t = 3.88), emphasizing the importance of advanced technologies in achieving efficiency, waste reduction, and green innovation.

Interaction plots were utilized to visualize the moderating effects (Dawson, 2014). Fig. 3 illustrates that higher organizational change capability strengthens the positive relationship between disruptive innovation and the circular economy. Similarly, Fig. 4 demonstrates that financial resilience enhances this relationship, with a more pronounced effect under higher financial resilience levels.

The model demonstrated moderate to substantial explanatory power, with an R-squared value value of 0.481 for the circular economy (Hair et al., 2017). Predictive validity was evaluated using holdout samples and cross-validation techniques, employing the PLS prediction algorithm in SmartPLS as suggested by Shmueli et al. (2019). Prediction statistics such as root mean squared error (RMSE), mean absolute error (MAE), and Q2 predict values confirmed the predictive performance of the PLS path model. These results are summarized in Table 5, which reports all RMSE, MAE, and Q2 values. Comparison with a linear regression model (LM) indicated that the PLS model exhibited lower prediction errors and higher Q2 values, further validating its predictive power (Hair et al., 2019).

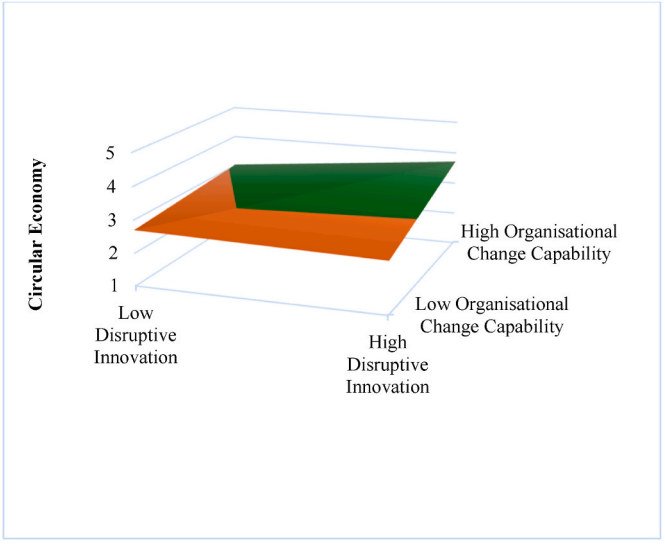



**Fig. 3.** Interaction plot of disruptive innovation and organisational change capability toward the circular economy.

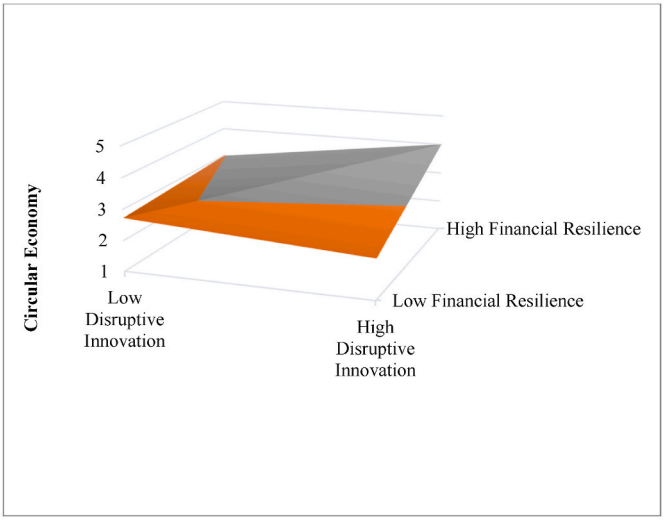



**Fig. 4.** Interaction plot of disruptive innovation and financial resilience toward the circular economy.

## 5. Discussion

This study explored the interplay among managerial and technical AI skills, disruptive innovation, and their impact on the circular economy, focusing on the technology and manufacturing sectors in Baghdad, Iraq. Drawing on dynamic capabilities theory (Teece et al., 1997), the research examined how organizations adapt, integrate, and reconfigure resources to navigate complex and volatile environments. The findings demonstrate that managerial and technical AI skills are pivotal drivers of disruptive innovation, enabling firms to transition toward circular economy practices that promote sustainability and long-term value creation. By leveraging AI competencies, organizations foster innovative solutions and enhance their ability to address pressing environmental and economic challenges.

The results are consistent with the notion that managerial AI skills empower leaders to integrate AI into business strategies, optimize processes, and identify opportunities (Chowdhury et al., 2023; Soto-Acosta, 2020, 2024). These skills also enhance organizational adaptability, enabling proactive responses to technological disruptions and fostering an environment conducive to innovation (Sousa & Rocha, 2019). Similarly, technical AI skills, such as expertise in machine learning and data analytics, play a critical role in enabling firms to design AI-driven solutions that disrupt traditional market structures and drive efficiency (Wael Al-Khatib, 2023). Together, these competencies reinforce the transformative potential of AI in achieving competitive advantages and sustainable growth.

The findings also align with the literature on disruptive innovation as a driver of circular economy practices. Disruptive innovation introduces technologies and business models that transform market dynamics and

**Table 5**
PLS-predict assessment.

| PLS-SEM | | | | LM | | | | PLS-LM | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Indicators | RMSE | MAE | $Q^2$ | Indicators | RMSE | MAE | $Q^2$ | RMSE | MAE | $Q^2$ |
| MAIS1 | 0.577 | 0.470 | 0.373 | MAIS1 | 0.608 | 0.517 | 0.303 | −0.031 | −0.047 | 0.070 |
| MAIS2 | 0.579 | 0.483 | 0.257 | MAIS2 | 0.591 | 0.496 | 0.224 | −0.012 | −0.013 | 0.033 |
| MAIS3 | 0.519 | 0.437 | 0.404 | MAIS3 | 0.598 | 0.518 | 0.210 | −0.079 | −0.081 | 0.194 |
| MAIS4 | 0.626 | 0.516 | 0.318 | MAIS4 | 0.649 | 0.552 | 0.269 | −0.023 | −0.036 | 0.049 |
| MAIS5 | 0.610 | 0.510 | 0.260 | MAIS5 | 0.634 | 0.546 | 0.199 | −0.024 | −0.036 | 0.061 |
| MAIS6 | 0.614 | 0.521 | 0.248 | MAIS6 | 0.649 | 0.556 | 0.159 | −0.035 | −0.035 | 0.089 |
| MAIS7 | 0.679 | 0.552 | 0.176 | MAIS7 | 0.718 | 0.583 | 0.079 | −0.039 | −0.031 | 0.097 |
| TAIS1 | 0.673 | 0.554 | 0.220 | TAIS1 | 0.705 | 0.574 | 0.144 | −0.032 | −0.020 | 0.076 |
| TAIS2 | 0.772 | 0.591 | 0.157 | TAIS2 | 0.809 | 0.619 | 0.074 | −0.037 | −0.028 | 0.083 |
| TAIS3 | 0.686 | 0.575 | 0.146 | TAIS3 | 0.692 | 0.581 | 0.132 | −0.006 | −0.006 | 0.014 |
| TAIS4 | 0.538 | 0.436 | 0.310 | TAIS4 | 0.556 | 0.477 | 0.264 | −0.018 | −0.041 | 0.046 |
| TAIS5 | 0.601 | 0.496 | 0.355 | TAIS5 | 0.644 | 0.537 | 0.260 | −0.043 | −0.041 | 0.095 |
| TAIS6 | 0.592 | 0.486 | 0.282 | TAIS6 | 0.621 | 0.525 | 0.210 | −0.029 | −0.039 | 0.072 |
| TAIS7 | 0.618 | 0.515 | 0.247 | TAIS7 | 0.649 | 0.562 | 0.169 | −0.031 | −0.047 | 0.078 |
| DIN1 | 0.675 | 0.523 | 0.207 | DIN1 | 0.711 | 0.542 | 0.118 | −0.036 | −0.019 | 0.089 |
| DIN2 | 0.552 | 0.447 | 0.209 | DIN2 | 0.569 | 0.482 | 0.158 | −0.017 | −0.035 | 0.051 |
| DIN3 | 0.610 | 0.505 | 0.383 | DIN3 | 0.669 | 0.562 | 0.259 | −0.059 | −0.057 | 0.124 |
| OCC1 | 0.740 | 0.598 | 0.263 | OCC1 | 0.774 | 0.641 | 0.194 | −0.034 | −0.043 | 0.069 |
| OCC2 | 0.609 | 0.508 | 0.342 | OCC2 | 0.646 | 0.547 | 0.260 | −0.037 | −0.039 | 0.082 |
| OCC3 | 0.711 | 0.563 | 0.368 | OCC3 | 0.740 | 0.592 | 0.316 | −0.029 | −0.029 | 0.052 |
| OCC4 | 0.624 | 0.508 | 0.296 | OCC4 | 0.633 | 0.521 | 0.277 | −0.009 | −0.013 | 0.019 |
| OCC5 | 0.687 | 0.528 | 0.328 | OCC5 | 0.752 | 0.584 | 0.196 | −0.065 | −0.056 | 0.132 |
| OCC6 | 0.612 | 0.514 | 0.341 | OCC6 | 0.644 | 0.523 | 0.271 | −0.032 | −0.009 | 0.070 |
| OCC7 | 0.519 | 0.437 | 0.404 | OCC7 | 0.598 | 0.518 | 0.210 | −0.079 | −0.081 | 0.194 |
| OCC8 | 0.954 | 0.778 | 0.248 | OCC8 | 0.055 | 0.878 | 0.081 | −0.101 | −0.100 | 0.167 |
| FR1 | 0.584 | 0.466 | 0.301 | RUL1 | 0.630 | 0.528 | 0.186 | −0.046 | −0.062 | 0.115 |
| FR2 | 0.622 | 0.513 | 0.352 | RUL2 | 0.644 | 0.548 | 0.305 | −0.022 | −0.035 | 0.047 |
| FR3 | 0.614 | 0.521 | 0.248 | INS1 | 0.648 | 0.555 | 0.160 | −0.034 | −0.034 | 0.088 |
| FR4 | 0.982 | 0.795 | 0.169 | INS2 | 0.042 | 0.875 | 0.065 | −0.060 | −0.080 | 0.104 |
| FR5 | 0.632 | 0.513 | 0.148 | INS3 | 0.664 | 0.550 | 0.060 | −0.032 | −0.037 | 0.088 |
| FR6 | 0.982 | 0.795 | 0.169 | EEB1 | 10.042 | 0.875 | 0.065 | −0.060 | −0.080 | 0.104 |
| FR7 | 0.632 | 0.513 | 0.148 | EEB2 | 0.664 | 0.550 | 0.060 | −0.032 | −0.037 | 0.088 |
| FR8 | 0.577 | 0.470 | 0.373 | EEB3 | 0.609 | 0.518 | 0.301 | −0.032 | −0.048 | 0.072 |
| FR9 | 0.610 | 0.510 | 0.260 | EEB4 | 0.635 | 0.546 | 0.198 | −0.025 | −0.036 | 0.062 |
| FR10 | 0.579 | 0.483 | 0.257 | EEB5 | 0.591 | 0.496 | 0.223 | −0.012 | −0.013 | 0.034 |
| CE1 | 0.983 | 0.829 | 0.173 | CE1 | 10.049 | 0.880 | 0.060 | −0.066 | −0.051 | 0.113 |
| CE2 | 0.626 | 0.516 | 0.318 | CE2 | 0.649 | 0.552 | 0.268 | −0.023 | −0.036 | 0.050 |
| CE3 | 0.685 | 0.569 | 0.183 | CE3 | 0.748 | 0.639 | 0.027 | −0.063 | −0.070 | 0.156 |
| CE4 | 0.632 | 0.513 | 0.148 | CE4 | 0.664 | 0.551 | 0.059 | −0.032 | −0.038 | 0.089 |
| CE5 | 0.584 | 0.466 | 0.301 | CE5 | 0.630 | 0.528 | 0.186 | −0.046 | −0.062 | 0.115 |
| CE6 | 0.982 | 0.795 | 0.169 | CE6 | 10.036 | 0.863 | 0.074 | −0.054 | −0.068 | 0.095 |
| CE7 | 0.697 | 0.571 | 0.249 | CE7 | 0.760 | 0.639 | 0.107 | −0.063 | −0.068 | 0.142 |
| CE8 | 0.577 | 0.470 | 0.373 | CE8 | 0.609 | 0.518 | 0.301 | −0.032 | −0.048 | 0.072 |
| CE9 | 0.610 | 0.510 | 0.260 | CE9 | 0.635 | 0.546 | 0.198 | −0.025 | −0.036 | 0.062 |
| CE10 | 0.579 | 0.483 | 0.257 | CE10 | 0.591 | 0.496 | 0.223 | −0.012 | −0.013 | 0.034 |

Notes: PLS-SEM= Partial Least Squares Structural Equation Modeling, LM = Linear Regression Model, RMSE = Root Mean Squared Error, MAE = Mean Absolute Error, $Q^2$ = $Q^2$ Predict.

address barriers to adopting circular economy principles, such as waste reduction and resource optimization (Liu et al., 2024; Saha et al., 2024). This study adds to the discourse by highlighting the enabling role of AI in advancing circular economy goals through innovative solutions and reimagined value chains.

The moderating effects of organizational change capability and financial resilience also emerged as critical factors in this relationship. Organizational change capability enables firms to effectively navigate the disruptions caused by technological advancements and sustainability transitions (Joel et al., 2024). It fosters a culture of adaptability and resilience, ensuring smoother adoption of innovative business models and circular practices (Shaik et al., 2024). Financial resilience further strengthens this dynamic by providing the stability needed to invest in innovation and sustainable practices, even under uncertain conditions (Miroshnychenko et al., 2024). Together, these factors highlight the importance of internal resources in amplifying the effects of disruptive innovation on circular economy outcomes.

The theoretical contributions of this study lie in its extension of dynamic capabilities theory (Teece et al., 1997), particularly by conceptualizing managerial and technical AI competencies as critical enablers of firms' adaptive capacity in disruptive and resource-constrained environments. While prior research has typically examined AI skills in isolation or within broader IT capabilities, this study addresses a specific knowledge gap by jointly analyzing managerial and technical AI skills and their interplay with disruptive innovation and circular economy outcomes. This integrated perspective advances understanding of how AI competencies function not merely as operational tools but as dynamic capabilities that facilitate strategic innovation and sustainable transformation. In doing so, the study challenges traditional static views of capability development by demonstrating how these skills contribute to the ongoing reconfiguration of resources and routines in response to environmental volatility. Moreover, by incorporating organizational change capability and financial resilience as moderating factors, the research adds theoretical nuance to the dynamic capabilities framework, emphasizing their role as amplifiers in the AI-disruption-sustainability nexus. These insights are especially relevant in underexplored contexts such as Baghdad's industrial sector, offering a contextually grounded extension of theory into emerging markets where uncertainty and constraints are pronounced.

Practically, the findings provide a strategic roadmap for

stakeholders. Businesses must prioritize the development of managerial and technical AI skills, as these are foundational for fostering innovation and maintaining competitiveness in an AI-driven landscape (Sullivan & Wamba, 2024). Policymakers can support this effort by incentivizing AI adoption and circular economy practices through targeted policies and investments in supportive infrastructure (Bansal et al., 2023). Investors are encouraged to recognize AI-driven innovation and sustainability as high-return opportunities, particularly when paired with strong financial and organizational resilience (Carmeli et al., 2020).

By aligning AI technologies with circular economy strategies, organizations, policymakers, and investors can collaboratively create a thriving, sustainable ecosystem. This approach aligns technological innovation with environmental responsibility, contributing to global sustainability goals while ensuring competitive advantage in an increasingly complex market landscape.

## 6. Conclusions, limitations, and future research

This study provides valuable perspectives on the role of AI capabilities in driving disruptive innovation and facilitating the transition to a circular economy. By examining the dynamic interplay between managerial and technical AI skills, disruptive innovation, and sustainability practices, the research contributes to a deeper understanding of how organizations can leverage AI to gain competitive advantages while addressing environmental and economic challenges.

Despite its contributions, the study has certain limitations. First, the research focused exclusively on the technology and manufacturing sectors in Baghdad, Iraq, which may not fully represent the diverse challenges and opportunities present in other industries or regions. The dynamics of AI adoption and circular economy practices could vary significantly across cultural, regulatory, and economic contexts. Additionally, the study's sample comprised general, IT, and operations managers, potentially excluding critical perspectives from other stakeholders, such as frontline employees, customers, or investors. Future research should address these limitations by exploring a wider range of industries and stakeholder groups to provide a more holistic view of AI's role in fostering disruptive innovation and sustainability across different contexts.

The methodology presents another limitation. The two-wave data collection approach offers valuable cross-sectional insights but may not fully capture the long-term effects of AI adoption on circular economy transitions. Longitudinal studies and mixed-method approaches could better shed light on how the integration of AI evolves over time and impacts sustainable business models.

Future research could also investigate novel variables to deepen our understanding of AI's potential in advancing sustainability. One promising avenue is AI-driven supply chain transparency, which could enable organizations to optimize resource use and minimize waste by tracking and managing product lifecycle data. Similarly, AI-enabled stakeholder engagement could enhance collaboration by helping businesses align their sustainability efforts with the needs and preferences of customers and suppliers. Exploring AI's application in product lifecycle management, including its ability to predict, design, and optimize reuse, repair, and recycling, could further support organizations in transitioning to circular economy models.

In addition to these variables, future studies could examine the moderating effects of collaborative innovation networks within and across industries. Such networks have the potential to foster knowledge sharing, resource pooling, and collective problem-solving, amplifying the benefits of AI-driven innovation for sustainability. Another critical area of investigation is digital resilience, which considers an organization's capacity to absorb technological disruptions and transform them into opportunities for building capabilities aligned with both innovation and sustainability goals.

In conclusion, this research underscores the transformative potential of AI in driving disruptive innovation and fostering a circular economy. Addressing the identified limitations and exploring the proposed avenues for future research will contribute to a more nuanced and comprehensive understanding of AI's role in shaping sustainable and competitive business ecosystems.

## CRediT authorship contribution statement

**Hussam Al Halbusi:** Writing – original draft, Methodology, Investigation, Conceptualization. **Simona Popa:** Writing – review & editing, Writing – original draft, Validation, Formal analysis, Data curation, Conceptualization. **Pedro Soto-Acosta:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Mohammad Alshallaqi:** Writing – original draft, Methodology, Investigation, Conceptualization.

## Data availability

Data will be made available on request.

## References


Ajayi, F. A., & Udeh, C. A. (2024). Review of workforce upskilling initiatives for emerging technologies in IT. *International Journal of Management & Entrepreneurship Research, 6*(4), 1119–1137.

Al Halbusi, H., Al-Sulaiti, K. I., Alalwan, A. A., & Al-Busaidi, A. S. (2025). AI capability and green innovation impact on sustainable performance: Moderating role of big data and knowledge management. *Technological Forecasting and Social Change, 210*, Article 123897.

Al Halbusi, H., Popa, S., Alshibani, S. M., & Soto-Acosta, P. (2024). Greening the future: Analyzing green entrepreneurial orientation, green knowledge management and digital transformation for sustainable innovation and circular economy. *European Journal of Innovation Management*. https://doi.org/10.1108/EJIM-022024-0169. ahead-of-print No. ahead-of-print.

Amankwah-Amoah, J., Abdalla, S., Mogaji, E., Elbanna, A., & Dwivedi, Y. K. (2024). The impending disruption of creative industries by generative AI: Opportunities, challenges, and research agenda. *International Journal of Information Management*, Article 102759.

An, M., Lin, J., & Luo, X. R. (2024). The impact of human AI skills on organizational innovation: The moderating role of digital organizational culture. *Journal of Business Research, 182*, Article 114786.

Arana-Landin, G., Sigüenza, W., Landeta-Manzano, B., & Laskurain-Iturbe, I. (2024). Circular economy: On the road to ISO 59000 family of standards. *Corporate Social Responsibility and Environmental Management, 31*(3), 1977–2009.

Arora, A., Gupta, S., Devi, C., & Walia, N. (2023). Customer experiences in the era of artificial intelligence (AI) in context to FinTech: A fuzzy AHP approach. *Benchmarking: An International Journal, 30*(10), 4342–4369.

Arranz, C. F., Sena, V., & Kwong, C. (2023). Dynamic capabilities and institutional complexity: exploring the impact of innovation and financial support policies on the circular economy. *IEEE Transactions on Engineering Management, 71*, 9966–9980.

Ayaz, O., & Tatoglu, E. (2024). Unveiling the power of social value: Catalyzing circular economy in emerging market SMEs. *Journal of Cleaner Production, 453*, Article 142245.

Baabdullah, A. M. (2024). The precursors of AI adoption in business: Towards an efficient decision-making and functional performance. *International Journal of Information Management, 75*, Article 102745.

Baldassarre, B., Schepers, M., Bocken, N., Cuppen, E., Korevaar, G., & Calabretta, G. (2019). Industrial symbiosis: Towards a design process for eco-industrial clusters by integrating circular economy and industrial ecology perspectives. *Journal of Cleaner Production, 216*, 446–460.

Bansal, A., Panchal, T., Jabeen, F., Mangla, S. K., & Singh, G. (2023). A study of human resource digital transformation (HRDT): A phenomenon of innovation capability led by digital and individual factors. *Journal of Business Research, 157*, Article 113611.

Bhatia, M. S., Kumar, S., Gangwani, K. K., & Kaur, B. (2024). Key capabilities for closed-loop supply chain: Empirical evidence from manufacturing firms. *Technological Forecasting and Social Change, 208*, Article 123717.

Brink, A., Benyayer, L. D., & Kupp, M. (2024). Decision-making in organizations: Should managers use AI? *Journal of Business Strategy, 45*(4), 267–274.

Brislin, R. W. (1980). Cross-cultural research methods: Strategies, problems, applications. In I. Altman, A. Rapoport, & J. F. Wohlwill (Eds.), *Human behavior and environment* (pp. 47–82). New York: Plenum Press.

Carmeli, A., Dothan, A., & Boojihawon, D. K. (2020). Resilience of sustainability-oriented and financially-driven organizations. *Business Strategy and the Environment, 29*(1), 154–169.

Castro-Lopez, A., Iglesias, V., & Santos-Vijande, M. L. (2023). Organizational capabilities and institutional pressures in the adoption of circular economy. *Journal of Business Research, 161*, Article 113823.

Cegarra-Navarro, J. G., Soto-Acosta, P., & Wensley, A. K. (2016). Structured knowledge processes and firm performance: The role of organizational agility. *Journal of Business Research, 69*(5), 1544–1549.

Chowdhury, S., Budhwar, P., Dey, P. K., Joel-Edgar, S., & Abadie, A. (2022). AI-employee collaboration and business performance: Integrating knowledge-based view, socio-technical systems and organisational socialisation framework. *Journal of Business Research, 144*, 31–49.

Chowdhury, S., Dey, P., Joel-Edgar, S., Bhattacharya, S., Rodriguez-Espindola, O., Abadie, A., & Truong, L. (2023). Unlocking the value of artificial intelligence in human resource management through AI capability framework. *Human Resource Management Review, 33*(1), Article 100899.

Dawson, J. F. (2014). Moderation in management research: What, why, when, and how. *Journal of Business and Psychology, 29*(1), 1–19. https://doi.org/10.1007/s10869-013-9308-7

Du, J., Zhu, S., & Li, W. H. (2023). Innovation through internationalization: A systematic review and research agenda. *Asia Pacific Journal of Management, 40*(3), 1217–1251.

Gallego-Schmid, A., Chen, H. M., Sharmina, M., & Mendoza, J. M. F. (2020). Links between circular economy and climate change mitigation in the built environment. *Journal of Cleaner Production, 260*, Article 121115.

Guo, H., & Tsinopoulos, C. (2023). Enabling a circular economy through green manufacturing in Chinese apparel manufacturers: Antecedents and outcomes. *IEEE Transactions on Engineering Management, 71*, 9540–9554.

Gupta, S., Modgil, S., Choi, T. M., Kumar, A., & Antony, J. (2023). Influences of artificial intelligence and blockchain technology on financial resilience of supply chains. *International Journal of Production Economics, 261*, Article 108868.

Hair, J. F., Hult, G. T. M., Ringle, C., & Sarstedt, M. (2017). *A primer on partial Least Squares structural equation modeling (PLS-SEM)* (2d ed.). London and Thousand Oaks, CA: Sage.

Hair, J. F., Sarstedt, M., & Ringle, C. M. (2019). Rethinking some of the rethinking of partial least squares. *European Journal of Marketing, 53*(4), 566–584.

Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science, 43*(1), 115–135. https://doi.org/10.1007/s11747-014-0403-8

Hossain, M., Park, S., Suchek, N., & Pansera, M. (2024). Circular economy: A review of review articles. *Business Strategy and the Environment, 33*(7), 7077–7099.

Joel, O. S., Oyewole, A. T., Odunaiya, O. G., & Soyombo, O. T. (2024). Navigating the digital transformation journey: Strategies for startup growth and innovation in the digital era. *International Journal of Management & Entrepreneurship Research, 6*(3), 697–706.

Karman, A., Prokop, V., & de Sousa Jabbour, A. B. L. (2024). Circular economy practices as a shield for the long-term organisational and network resilience during crisis: Insights from an industrial symbiosis. *Journal of Cleaner Production*, Article 142822.

Khan, E. A., Chowdhury, M. M. H., Hossain, M. A., Mahmud, A. S., Baabdullah, A. M., & Dwivedi, Y. K. (2024). Performance of small firms in a circular economy: Configuring challenges and entrepreneurial orientation. *Management Decision, 62*(8), 2428–2450.

Kraus, S., Vonmetz, K., Orlandi, L. B., Zardini, A., & Rossignoli, C. (2023). Digital entrepreneurship: The role of entrepreneurial orientation and digitalization for disruptive innovation. *Technological Forecasting and Social Change, 193*, Article 122638.

Kuhlmann, M., Bening, C. R., & Hoffmann, V. H. (2023). How incumbents realize disruptive circular innovation-Overcoming the innovator's dilemma for a circular economy. *Business Strategy and the Environment, 32*(3), 1106–1121.

Kutaula, S., Gillani, A., Leonidou, L. C., & Christodoulides, P. (2022). Integrating fair trade with circular economy: Personality traits, consumer engagement, and ethically-minded behavior. *Journal of Business Research, 144*, 1087–1102.

Liu, Y., Xing, Y., Vendrell-Herrero, F., & Bustinza, O. F. (2024). Setting contextual conditions to resolve grand challenges through responsible innovation: A comparative patent analysis in the circular economy. *Journal of Product Innovation Management, 41*(2), 449–472.

Lu, J., Rodenburg, K., Foti, L., & Pegoraro, A. (2022). Are firms with better sustainability performance more resilient during crises? *Business Strategy and the Environment, 31*(7), 3354–3370.

Memon, M. A., Ting, H., Ramayah, T., Chuah, F., & Cheah, J. H. (2017). A review of the methodological misconceptions and guidelines related to the application of structural equation modeling: A Malaysian scenario. *Journal of Applied Structural Equation Modeling, 1*(1), 1–13.

Mikalef, P., & Gupta, M. (2021). Artificial intelligence capability: Conceptualization, measurement calibration, and empirical study on its impact on organizational creativity and firm performance. *Information & Management, 58*(3), Article 103434.

Milios, L., Beqiri, B., Whalen, K. A., & Jelonek, S. H. (2019). Sailing towards a circular economy: Conditions for increased reuse and remanufacturing in the Scandinavian maritime sector. *Journal of Cleaner Production, 225*, 227–235.

Miroshnychenko, I., Vocalelli, G., De Massis, A., Grassi, S., & Ravazzolo, F. (2024). The COVID-19 pandemic and family business performance. *Small Business Economics, 62*(1), 213–241.

Mishra, A., Dutta, P., Jayasankar, S., Jain, P., & Mathiyazhagan, K. (2023). A review of reverse logistics and closed-loop supply chains in the perspective of circular economy. *Benchmarking: An International Journal, 30*(3), 975–1020.

Olan, F., Arakpogun, E. O., Suklan, J., Nakpodia, F., Damij, N., & Jayawickrama, U. (2022). Artificial intelligence and knowledge sharing: Contributing factors to organizational performance. *Journal of Business Research, 145*, 605–615.

Pal, R., Torstensson, H., & Mattila, H. (2014). Antecedents of organizational resilience in economic crises—an empirical study of Swedish textile and clothing SMEs. *International Journal of Production Economics, 147*, 410–428.

Parise, S., Guinan, P. J., & Kafka, R. (2016). Solving the crisis of immediacy: How digital technology can transform the customer experience. *Business Horizons, 59*(4), 411–420.

Pascucci, F., Pizzichini, L., Sabatini, A., Temperini, V., & Mueller, J. (2024). Knowledge-based dynamic capabilities for managing paradoxical tensions in circular business model innovation: An empirical exploration of an incumbent firm. *Journal of Knowledge Management, 28*(11), 255–282.

Patel, A., & Singh, S. (2023). Implementing circular economy strategies in the automobile industry–a step toward creating sustainable automobiles. *Benchmarking: An International Journal, 30*(7), 2225–2233.

Podsakoff, P. M., Podsakoff, N. P., Williams, L. J., Huang, C., & Yang, J. (2024). Common method bias: It's bad, it's complex, it's widespread, and it's not easy to fix. *Annual Review of Organizational Psychology and Organizational Behavior, 11*(1), 17–61.

Popa, S., Soto-Acosta, P., & Palacios-Marqués, D. (2022). A discriminant analysis of high and low-innovative firms: The role of IT, human resources, innovation strategy, intellectual capital and environmental dynamism. *Journal of Knowledge Management, 26*(6), 1615–1632.

Purkayastha, A., & Gupta, V. K. (2023). Business group affiliation and entrepreneurial orientation: Contingent effect of level of internationalization and firm's performance. *Asia Pacific Journal of Management, 40*(2), 847–876.

Ringle, C. M., Wende, S., & Becker, J.-M. (2015). SmartPLS 4. Bonningstedt: SmartPLS. Retrieved November 30, 2024, from http://www.smartpls.com.

Rotjanakorn, A., Sadangharn, P., & Na-Nan, K. (2020). Development of dynamic capabilities for automotive industry performance under disruptive innovation. *Journal of Open Innovation: Technology, Market, and Complexity, 6*(4), 97.

Roux, M., Chowdhury, S., Kumar Dey, P., Vann Yaroson, E., Pereira, V., & Abadie, A. (2023). Small and medium-sized enterprises as technology innovation intermediaries in sustainable business ecosystem: Interplay between AI adoption, low carbon management and resilience. *Annals of Operations Research*, 1–50.

Saha, K., Dey, P. K., & Kumar, V. (2024). A comprehensive review of circular economy research in the textile and clothing industry. *Journal of Cleaner Production*, Article 141252.

Sahoo, S. (2024). Assessing the impact of stakeholder pressure and green data analytics on firm's environmental performance–understanding the role of green knowledge management and green technological innovativeness. *R & D Management, 54*(1), 3–20.

Saurabh, K., Rani, N., & Upadhyay, P. (2022). Towards blockchain led decentralized autonomous organization (DAO) business model innovations. *Benchmarking: An International Journal, 30*(2), 475–502.

Schindler, J., Kallmuenzer, A., & Valeri, M. (2024). Entrepreneurial culture and disruptive innovation in established firms–how to handle ambidexterity. *Business Process Management Journal, 30*(2), 366–387.

Scuotto, V., Magni, D., Palladino, R., & Nicotra, M. (2022). Triggering disruptive technology absorptive capacity by CIOs. Explorative research on a micro-foundation lens. *Technological Forecasting and Social Change, 174*, Article 121234.

Sehnem, S., Provensi, T., da Silva, T. H. H., & Pereira, S. C. F. (2022). Disruptive innovation and circularity in start-ups: A path to sustainable development. *Business Strategy and the Environment, 31*(4), 1292–1307.

Shaik, A. S., Alshibani, S. M., Jain, G., Gupta, B., & Mehrotra, A. (2024). Artificial intelligence (AI)-driven strategic business model innovations in small-and medium-sized enterprises. Insights on technological and strategic enablers for carbon neutral businesses. *Business Strategy and the Environment, 33*(4), 2731–2751.

Shmueli, G., Sarstedt, M., Hair, J., Cheah, J., Ting, H., Vaithilingam, S., & Ringle, C. (2019). Predictive model assessment in PLS-SEM: Guidelines for using PLSpredict. *European Journal of Marketing, 53*(11), 2322–2347.

Soto-Acosta, P. (2020). COVID-19 pandemic: Shifting digital transformation to a high-speed gear. *Information Systems Management, 37*(4), 260–266.

Soto-Acosta, P. (2024). Navigating uncertainty: Post-Pandemic issues on digital transformation. *Information Systems Management, 41*(1), 20–26.

Sousa, M. J., & Rocha, Á. (2019). Skills for disruptive digital business. *Journal of Business Research, 94*, 257–263.

Sullivan, Y., & Wamba, S. F. (2024). Artificial intelligence and adaptive response to market changes: A strategy to enhance firm performance and innovation. *Journal of Business Research, 174*, Article 114500.

Supriharyanti, E., & Sukoco, B. M. (2023). Organizational change capability: A systematic review and future research directions. *Management Research Review, 46*(1), 46–81.

Teece, D. J., Pisano, G., & Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal, 18*(7), 509–533.

Tiwari, P. (2023). Effect of innovation practices of banks on customer loyalty: An SEM-ANN approach. *Benchmarking: An International Journal, 30*(10), 4536–4568.

Tiwari, S., Bahuguna, P. C., & Srivastava, R. (2023). Smart manufacturing and sustainability: A bibliometric analysis. *Benchmarking: An International Journal, 30*(9), 3281–3301.

Vann Yaroson, E., Chowdhury, S., Mangla, S. K., Dey, P., Chan, F. T., & Roux, M. (2024). A systematic literature review exploring and linking circular economy and sustainable development goals in the past three decades (1991–2022). *International Journal of Production Research, 62*(4), 1399–1433.

wael AL-khatib, A. (2023). Drivers of generative artificial intelligence to fostering exploitative and exploratory innovation: A TOE framework. *Technology in Society, 75*, Article 102403.

Zahedi, J., Salehi, M., & Moradi, M. (2023). Identifying and classifying the financial resilience measurement indices using intuitive fuzzy DEMATEL. *Benchmarking: An International Journal, 30*(4), 1300–1321.

Zhang, F., Zhang, H., Brown, D. H., & Yin, X. (2023). Innovation and performance of manufacturing firms in aspirant markets: An institutional environment approach. *Asia Pacific Journal of Management, 40*(2), 435–482.